

# Lab: Securing Web Applications with HTTPS and Data & Configuration Archiving

**Objective:** Securing an application with HTTPS and archiving the application data, and configuration

**Tools required:** XAMPP

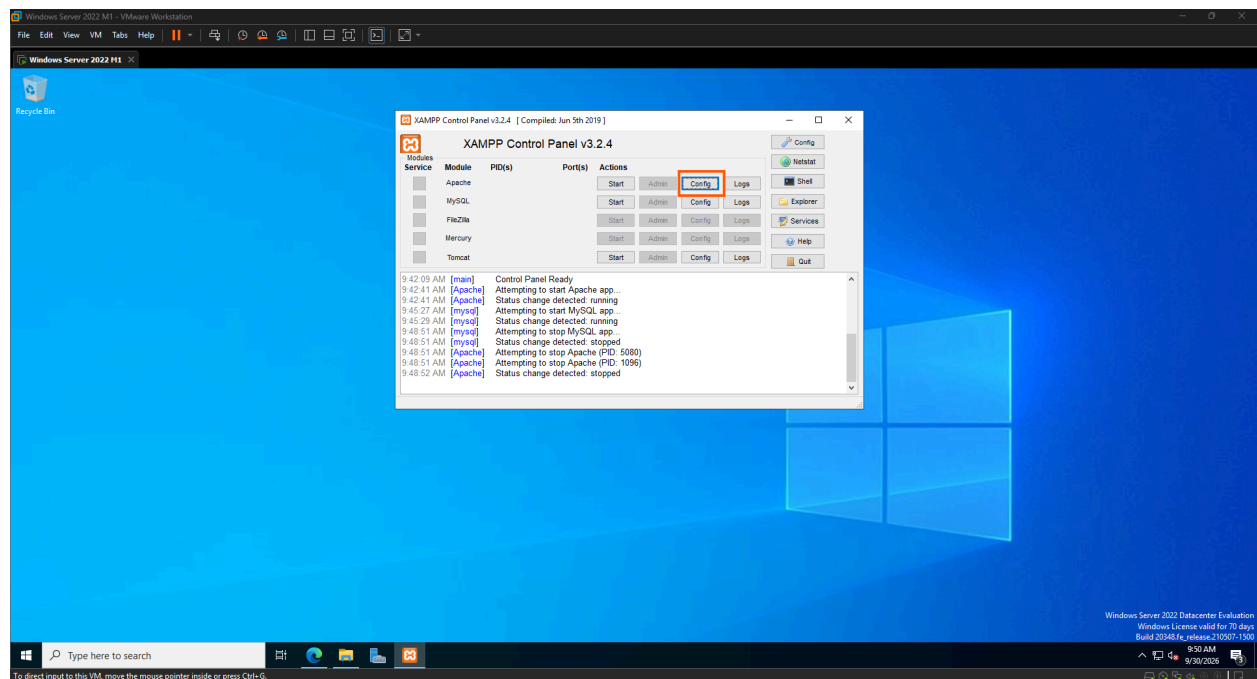
**Prerequisites:** Hosting of a webapp on XAMPP must be completed

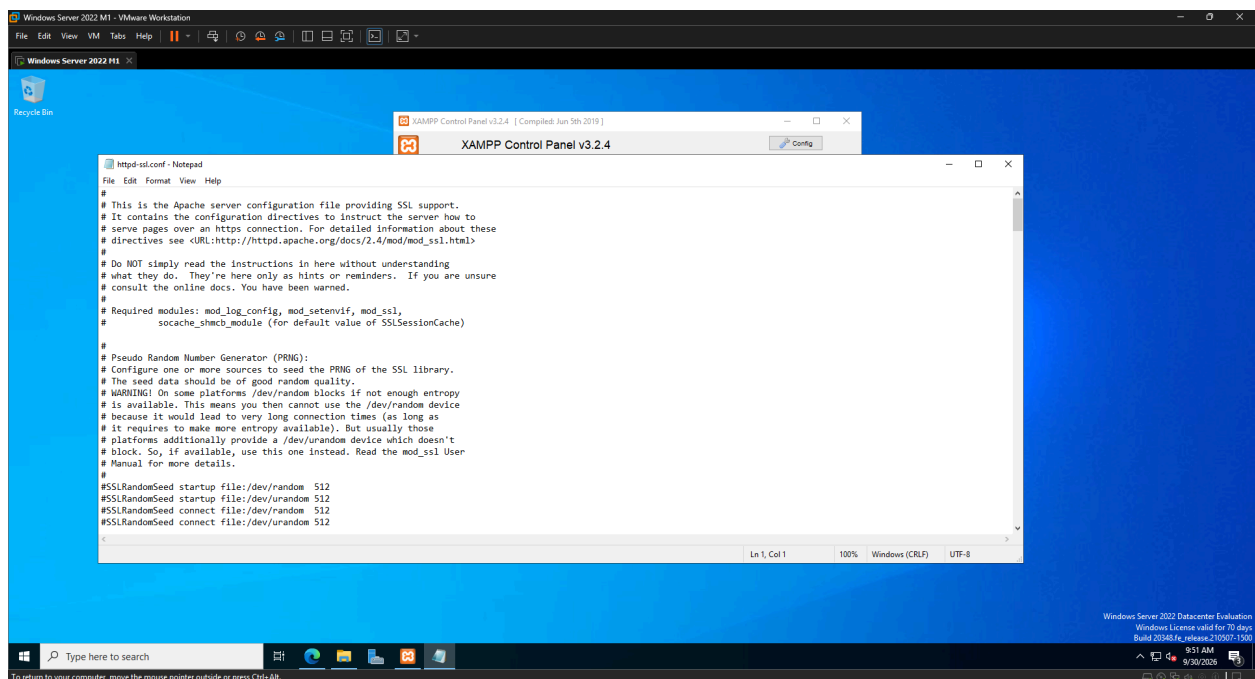
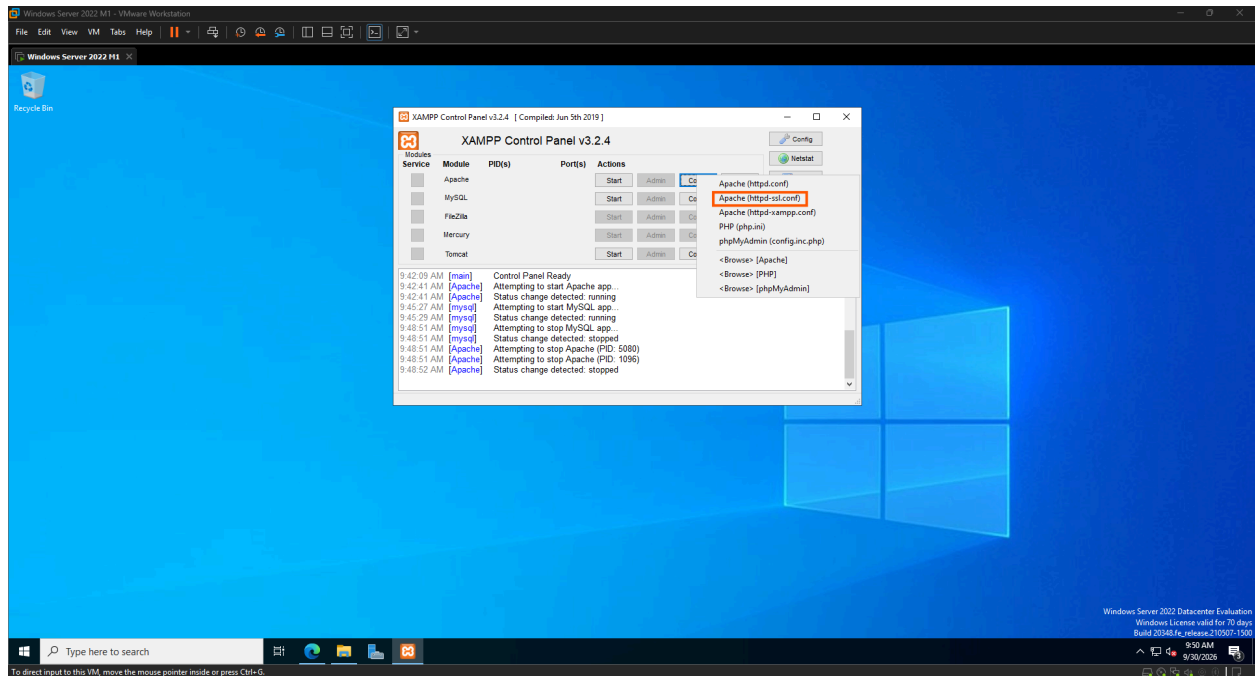
**Steps to be followed:**

1. Securing an application with HTTPS
2. Archiving the application data, and configuration

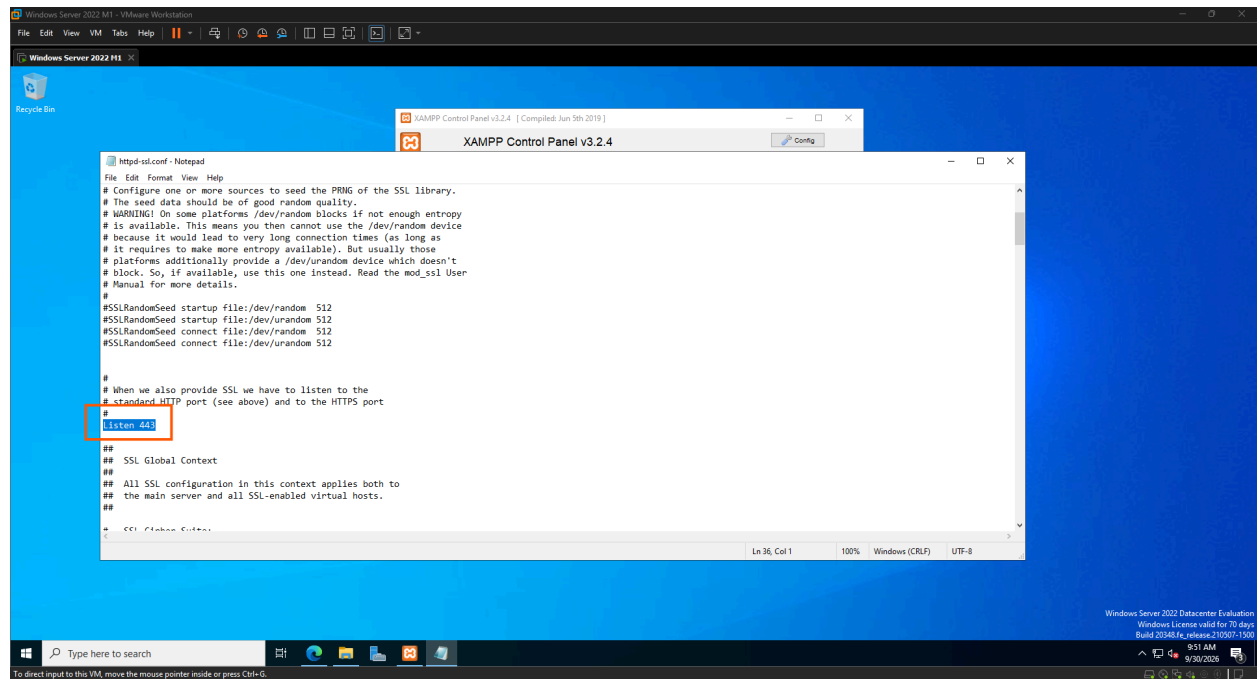
## Step 1: Securing an application with HTTPS

1.1 Launch Windows Server 2022 VM, start XAMPP, click on config & select Apache (httpd-ssl.conf)

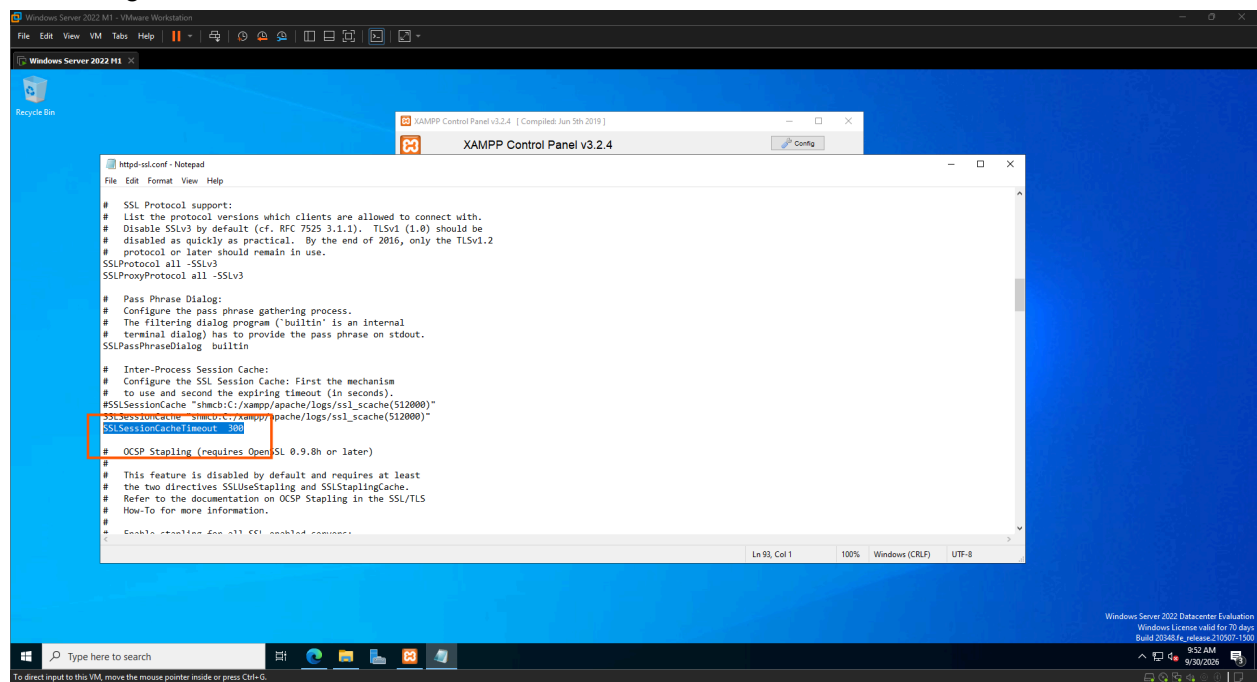


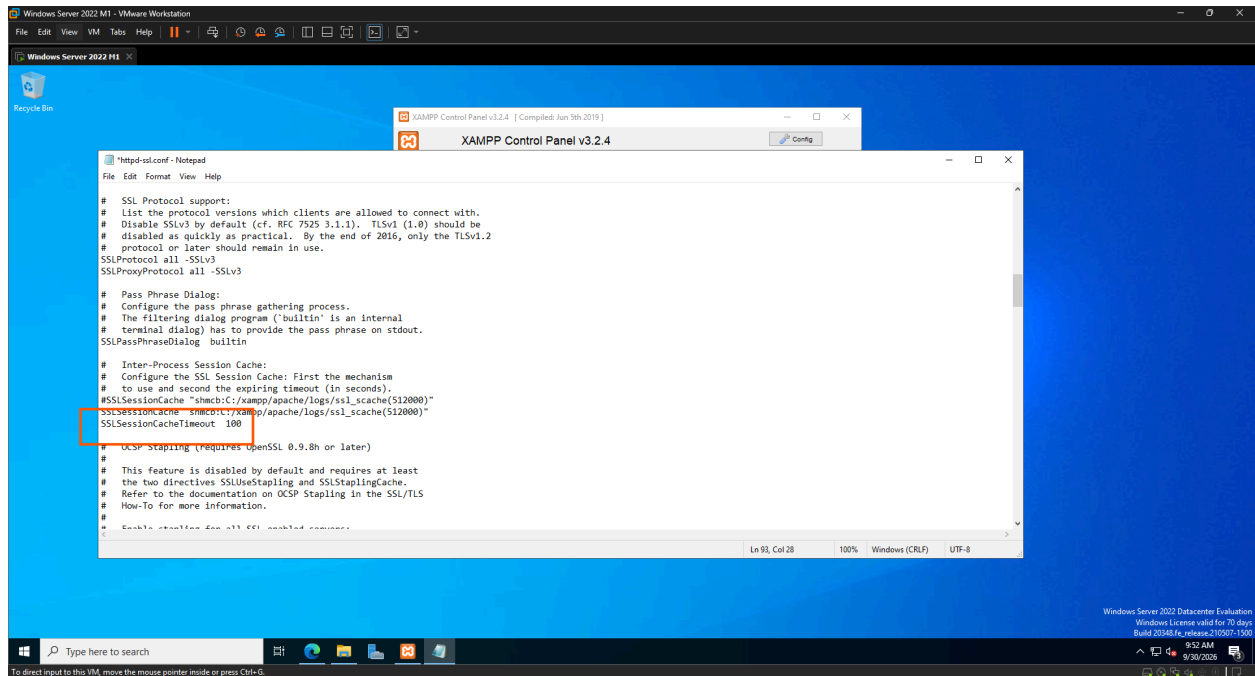


## 1.2 Make sure, It should listen on PORT 443

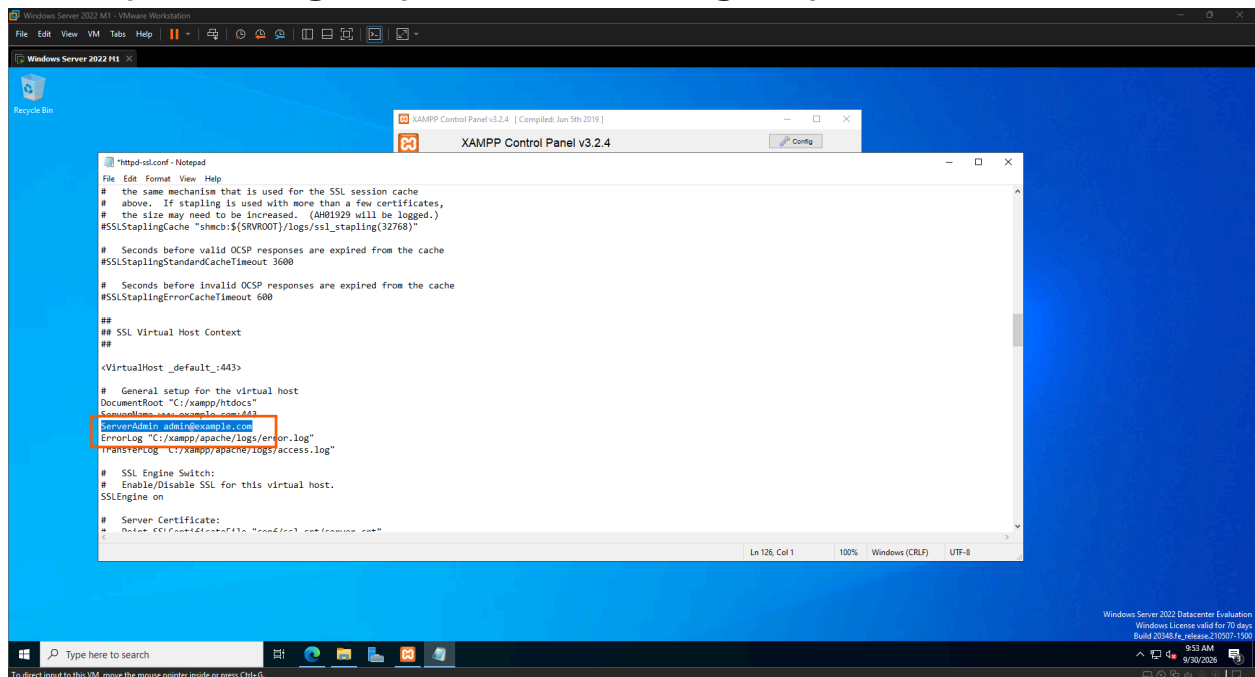


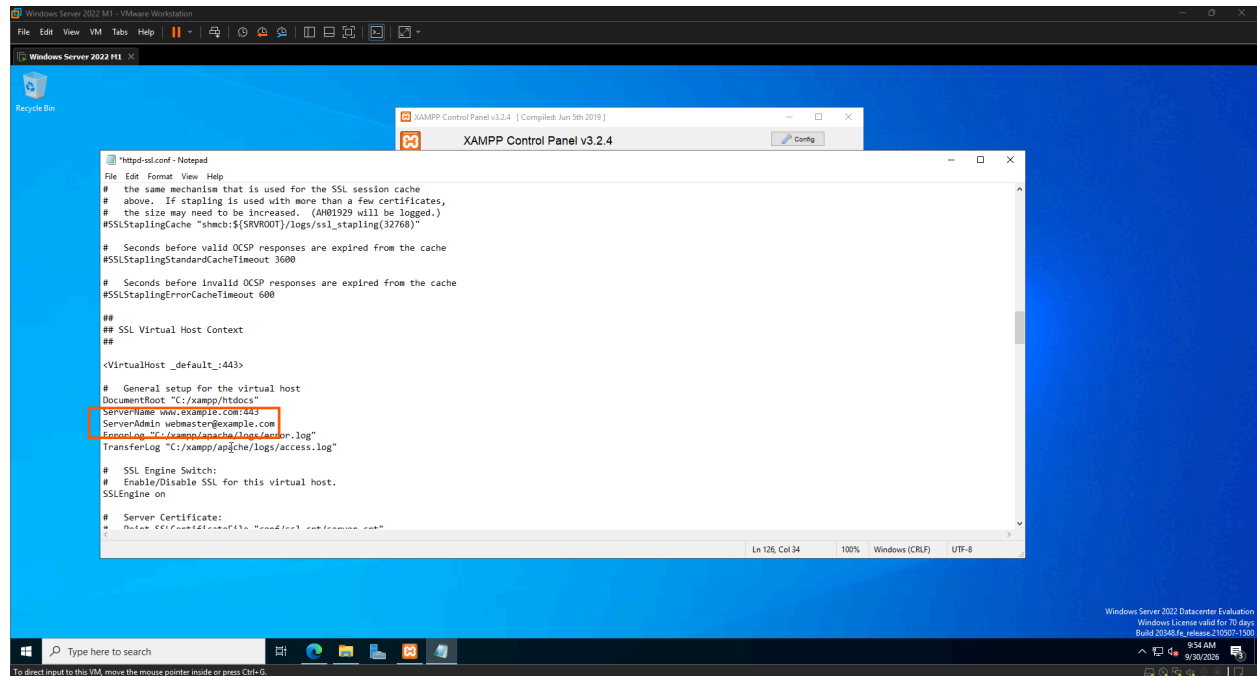
## 1.3 Change the “*SSLSessionCacheTimeout 300*” to “*SSLSessionCacheTimeout 100*”



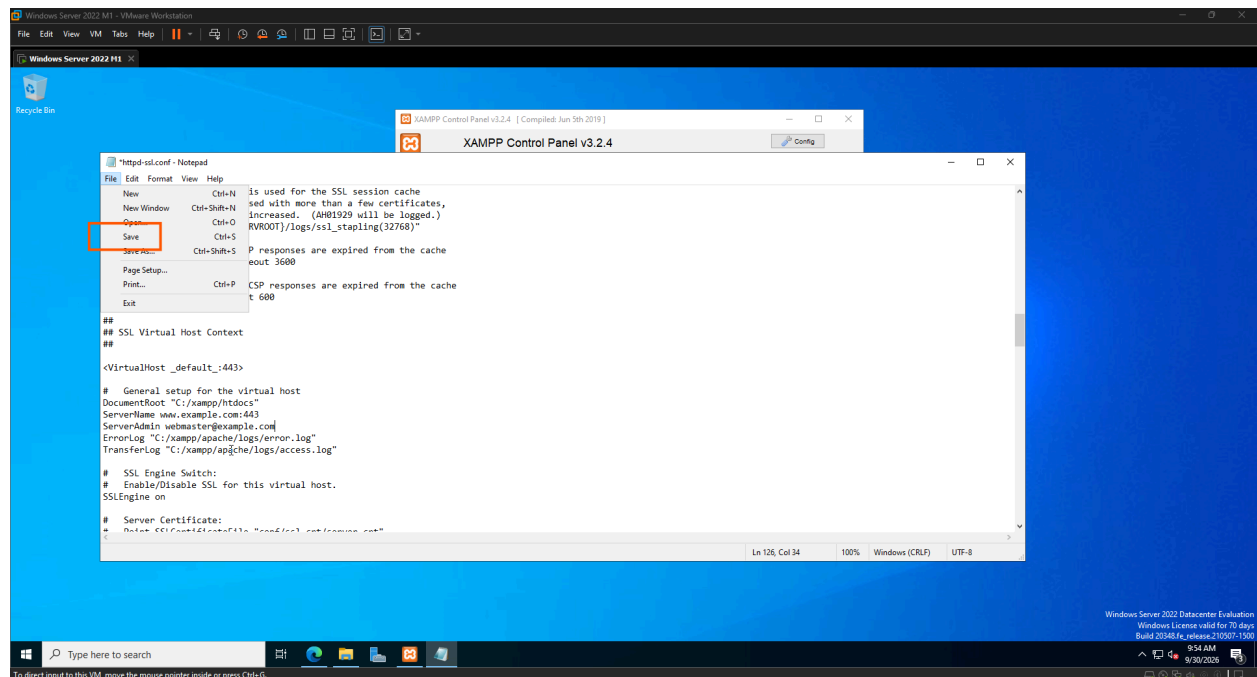


## 1.4 Change the *admin@example.com* to *webmaster@example.com*

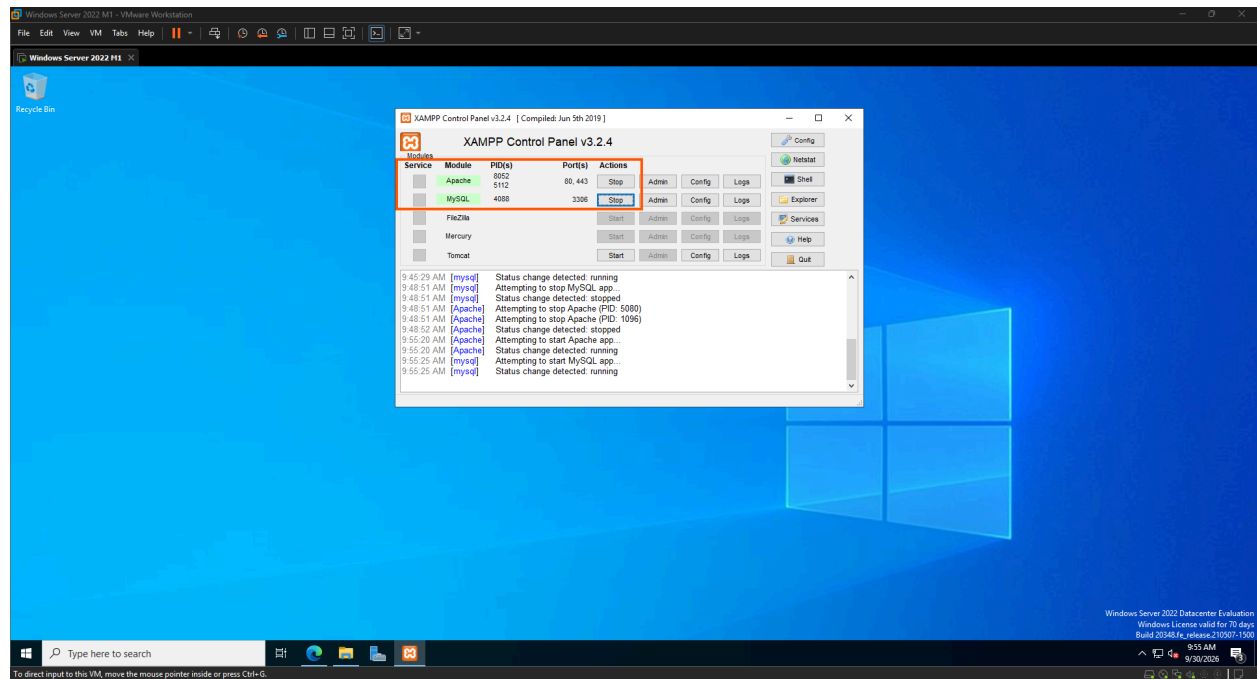




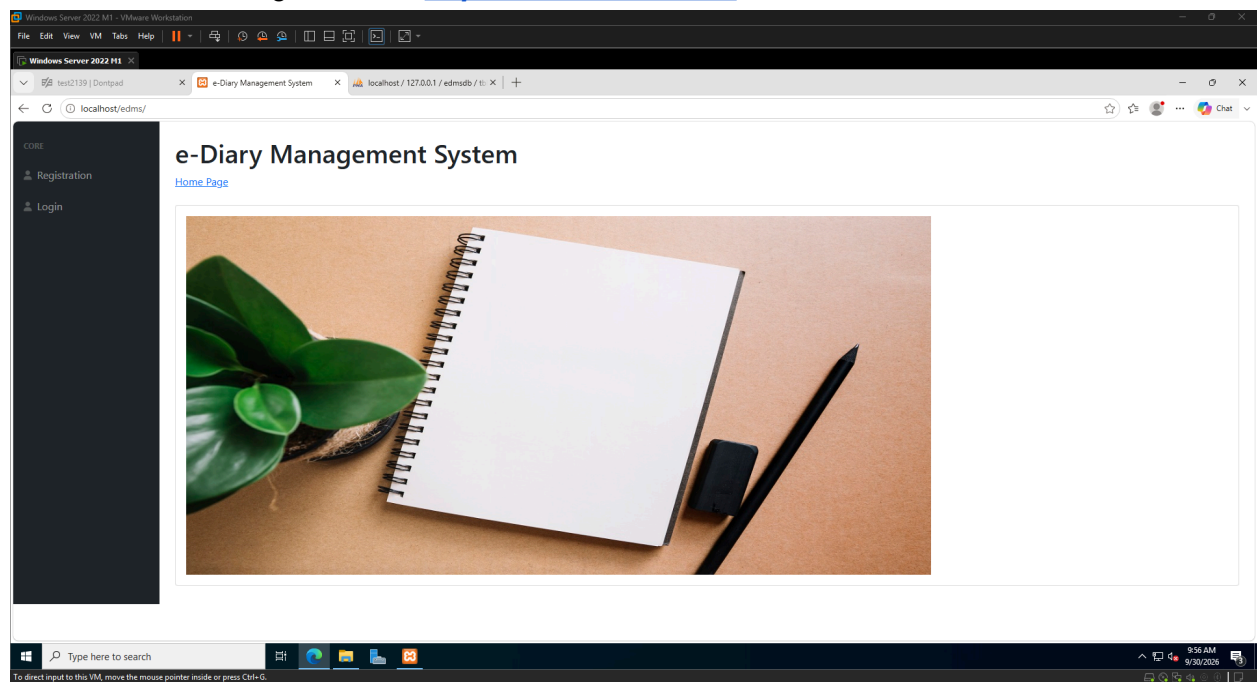
## 1.5 Save the file



## 1.6 Start the *Apache* and *MySQL*

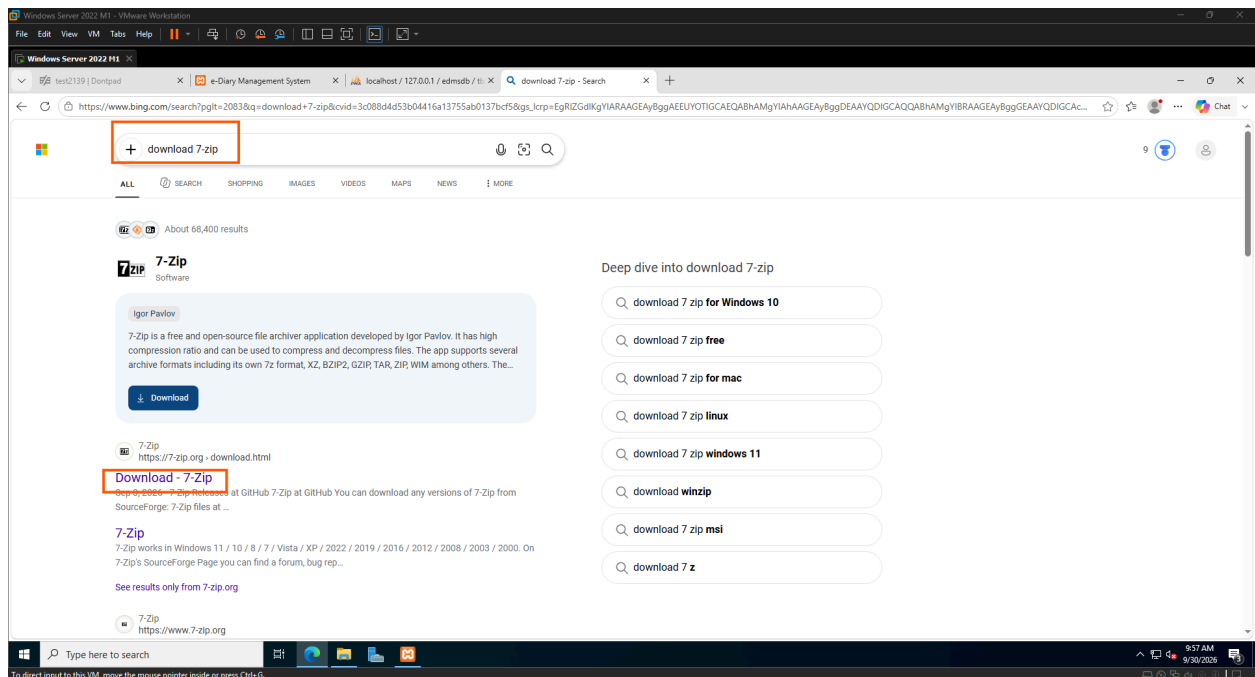


## 1.7 Search in the Edge browser <https://localhost/edms>

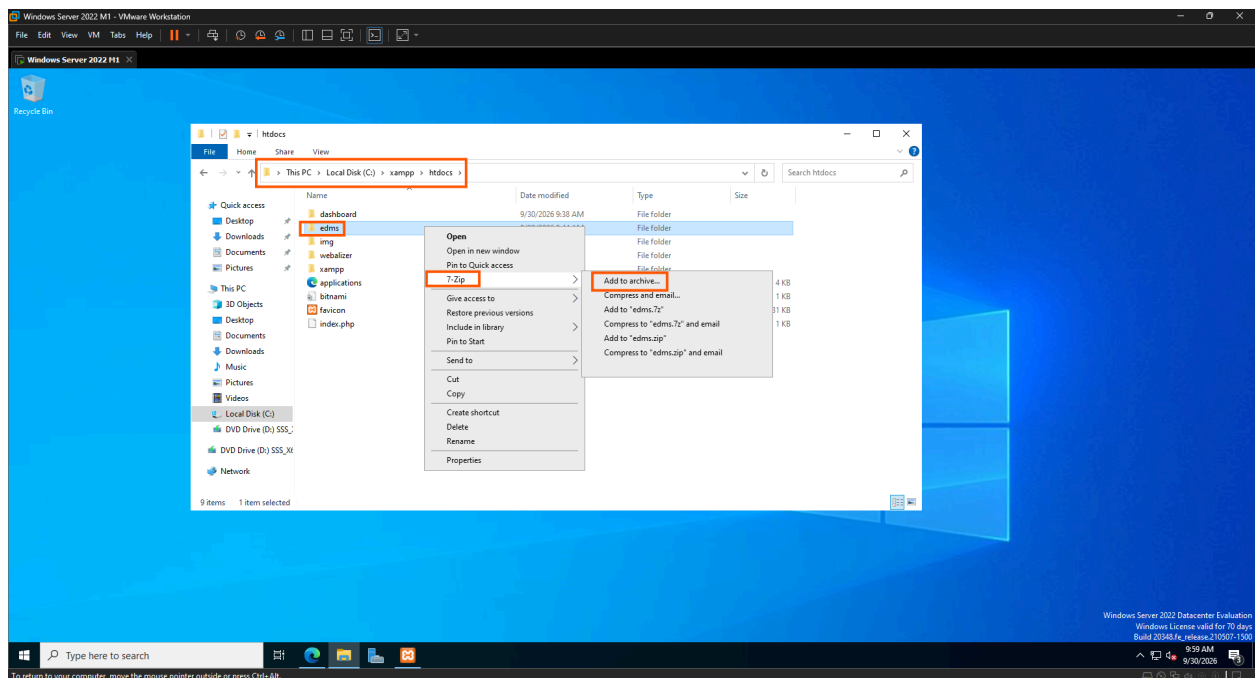


## Step 2: Archiving the application data, and configuration

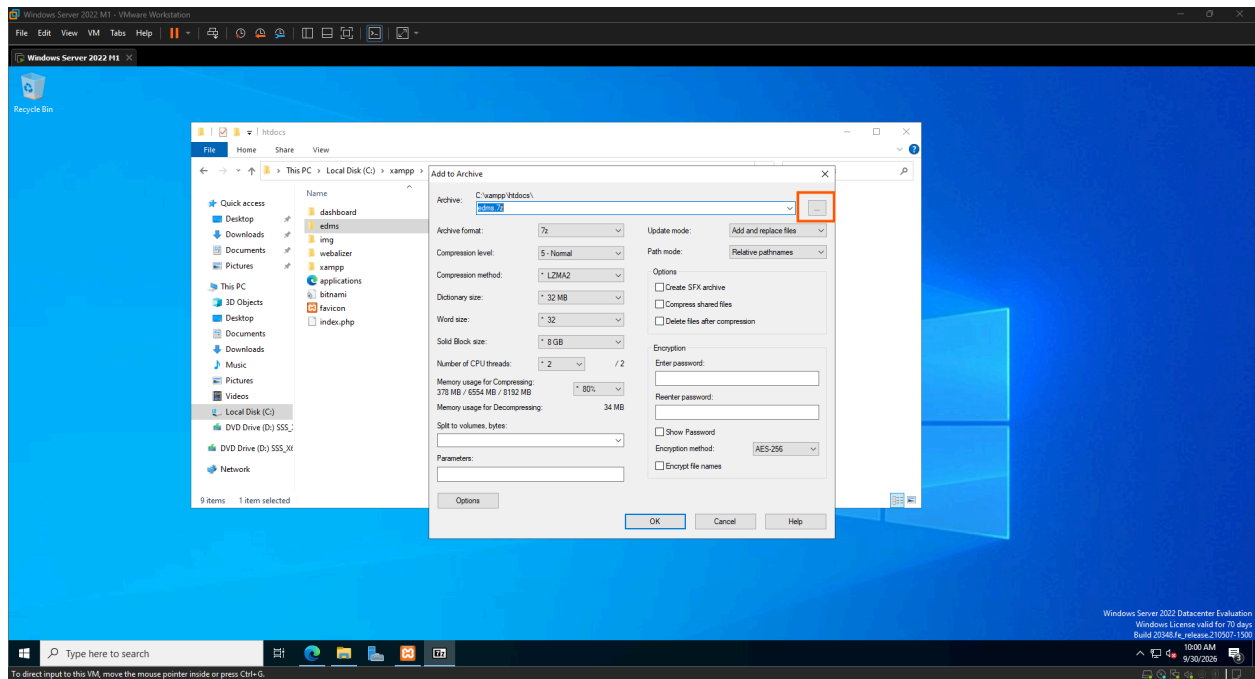
### 2.1 Open the **Edge browser** and search **7-zip** then click on Download & install it



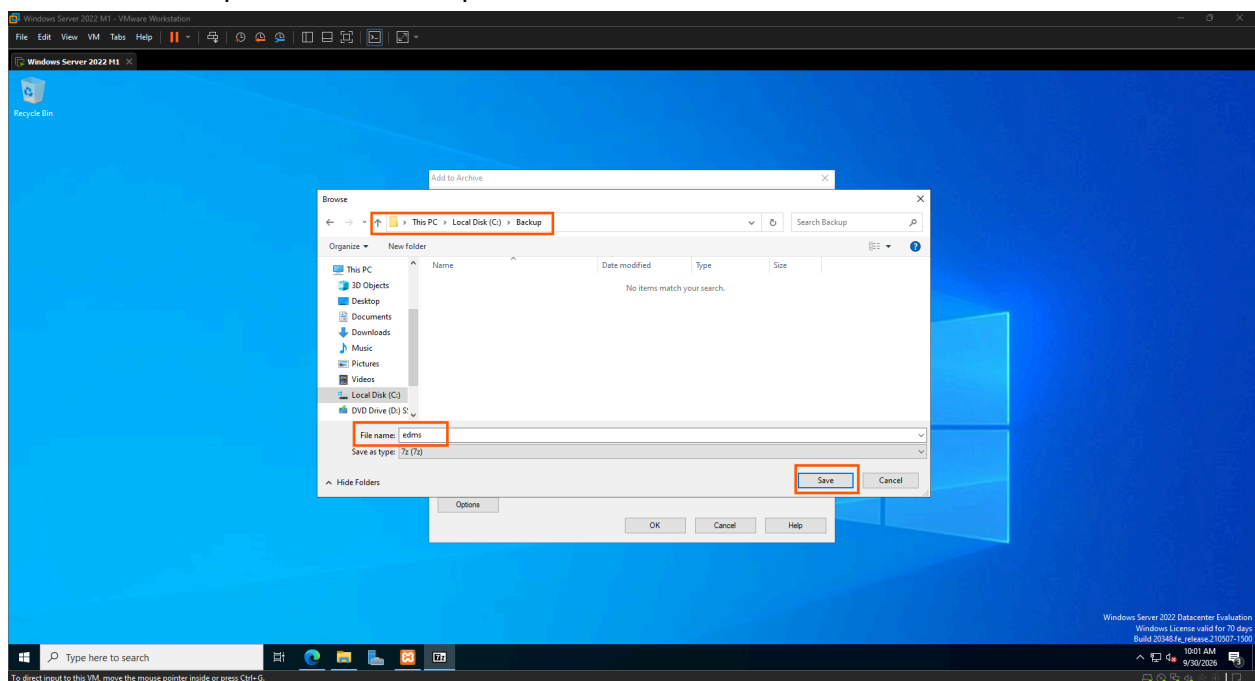
### 2.2 Now navigate to **C:\xampp\htdocs**, Right-click on the **edms** and select the **7-Zip** then click on **Add to archive...**

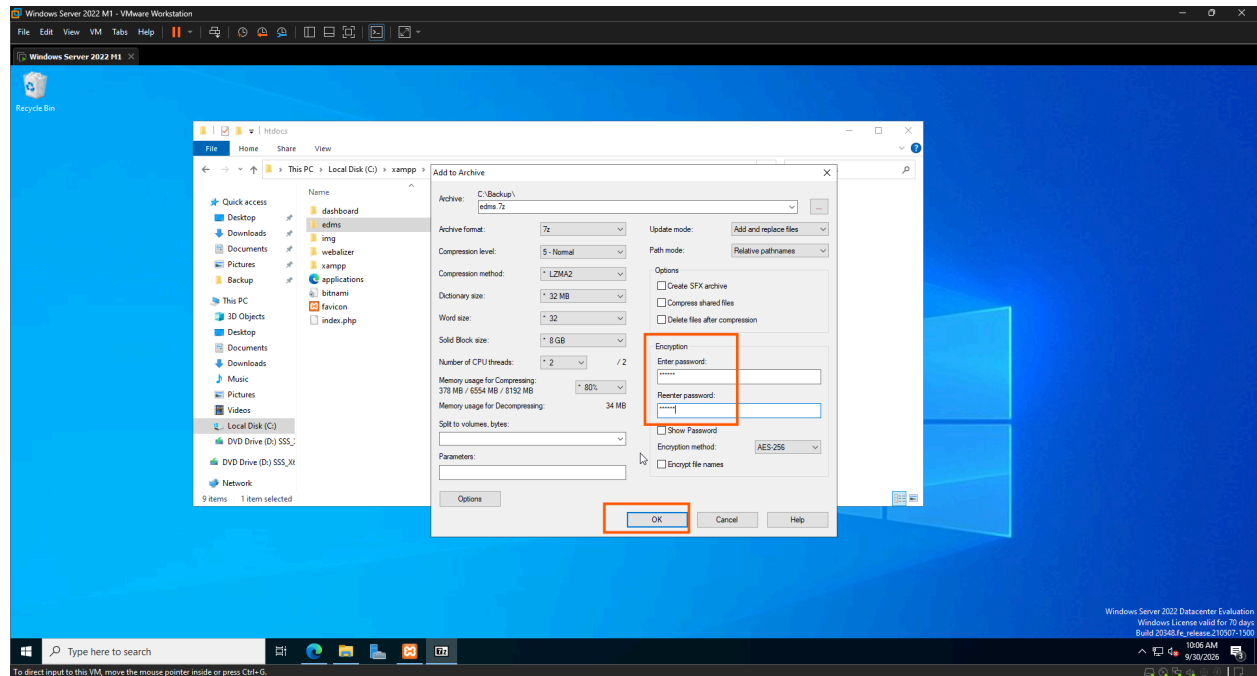


2.3 Click on the **3-dot** located at **top right** side.

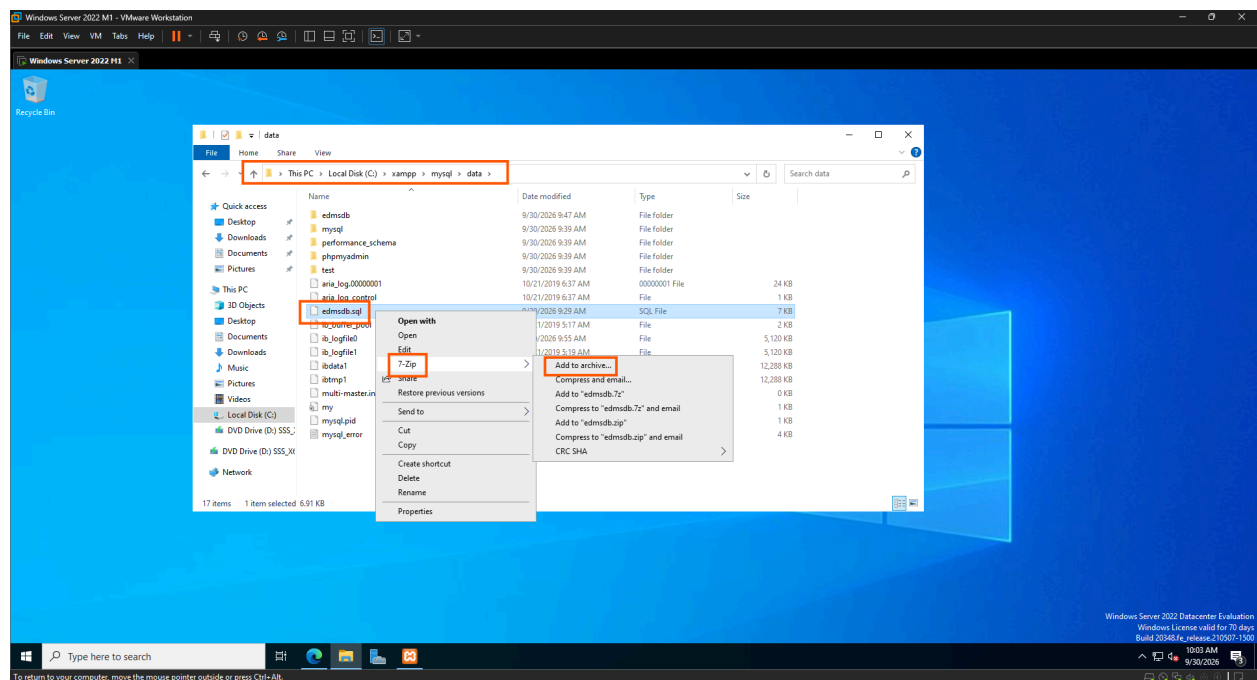


2.4 Navigate to the **C drive**, click on **New folder**, create a new folder, and name it **Backup** & save the edms zip file inside it with password.

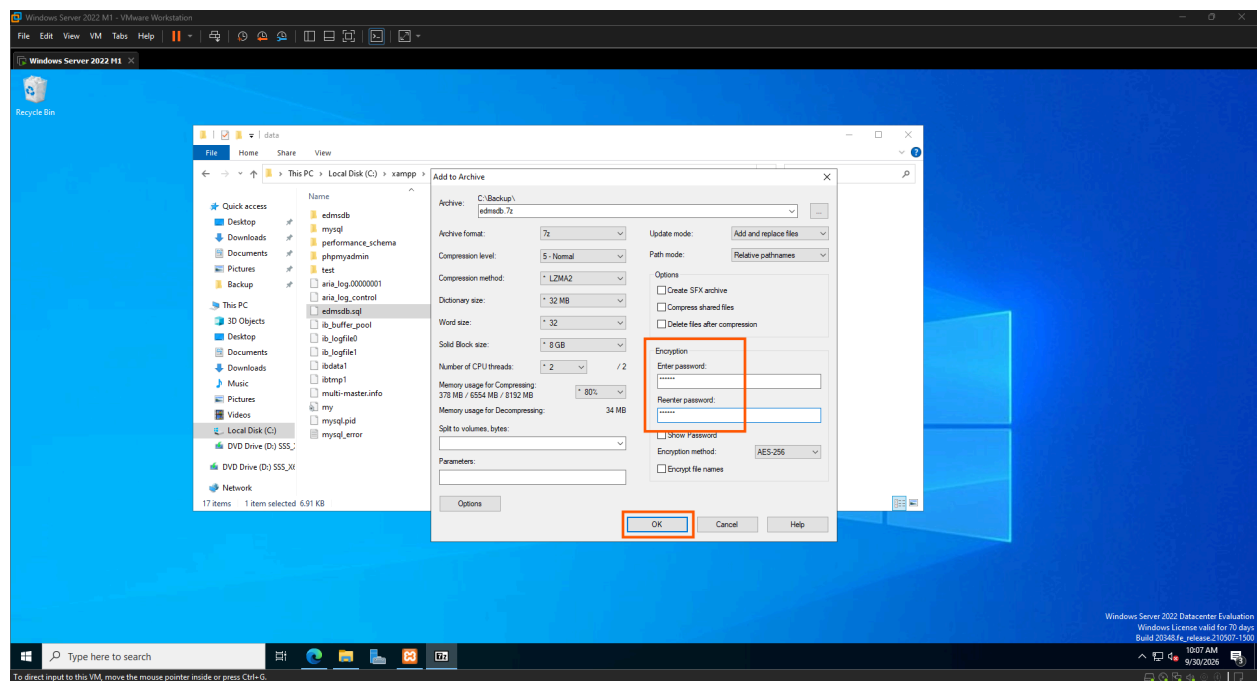
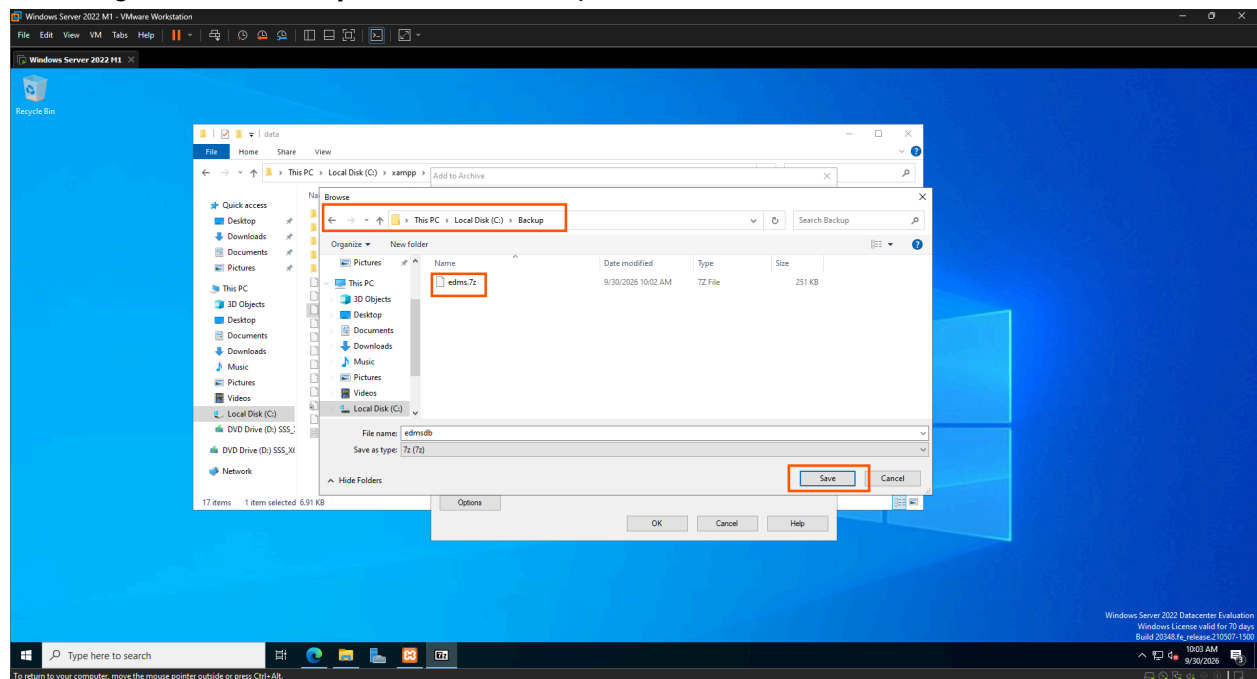




2.5 Now navigate to **C:\xampp\mysql\data**, Right-click on the **edmsdb.sql** and select **7-zip** then click on **Add to archive...**



## 2.6 Navigate to **C:\Backup** and **save** it with password.



## Conclusion:

Securing an application with HTTPS ensures encrypted communication between the server and clients, protecting sensitive data from interception and tampering. This is a critical step in building trust with users and adhering to security best practices. Archiving application data and configurations adds another layer of protection by preserving critical information for future reference, disaster recovery, or regulatory compliance. Together, these practices enhance the security, reliability, and maintainability of an application, fostering a robust and trustworthy system.

